

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1510

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 25

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Code of Conduct

I Prabhavathi R (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Prabhavathi R

CSA1510

Cloud Computing

CO1 - Assessment tool 1

Prabhavathi. R

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B.Tech IT

① Multi-cloud and Hybrid cloud strategies for Enterprise IT

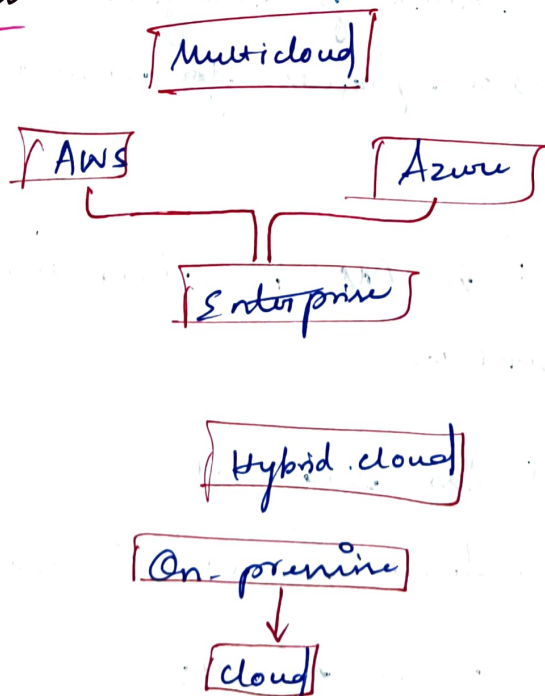
Deployment models

* Multi-cloud → using multiple cloud providers

Ceg: AWS + Azure

* Hybrid cloud → combination of on-premise + cloud

Diagram



Business Benefits

- * Avoid vendor lock-in
- * High Availability
- * Cost Optimization
- * Flexibility

Challenges

- * Complex management
- * Security risks
- * Data Integration Issues

Example

* A bank uses:

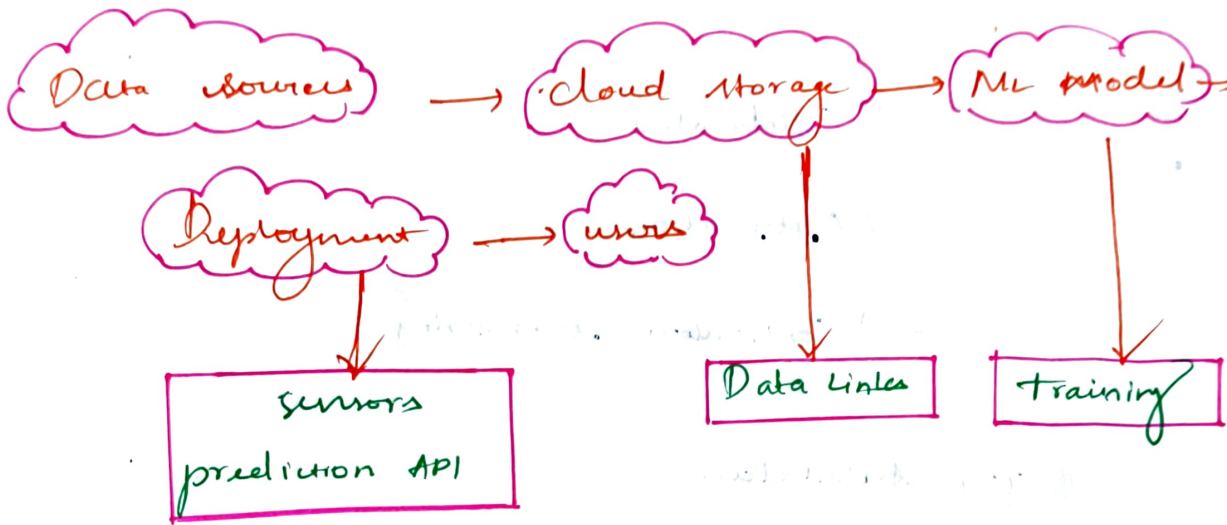
- AWS for storage
- Azure for Analytics
- on-premise for sensitive data

Role of AI & ML in cloud platforms cloud

AI/ML services

- AWS SageMaker
- Azure Machine Learning
- Google AI platform

Architecture



Benefits

- stability
- Faster model training
- pay-as-you-use

Applications

- * Healthcare - Disease prediction
- * Banking - Fraud detection
- * E-commerce - Recommendation systems

Example

- Amazon uses ML to recommend products.

③ Devops and CI/CD in cloud computing

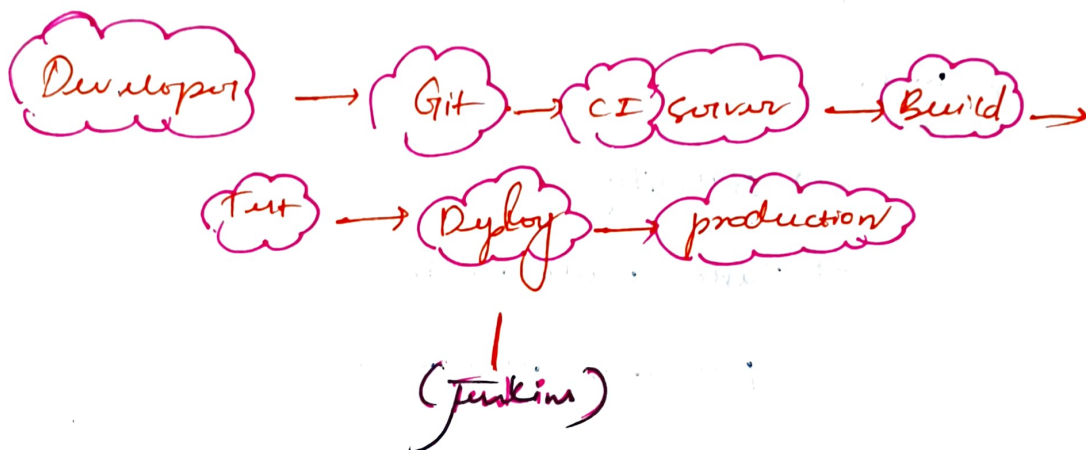
Devops principles

→ Collaborations

→ Automation

→ Continuous monitoring

CI/CD Architecture



CI (continuous Integration)

→ code merged frequently

→ Automated testing

CD (continuous Deployment)

→ Automatic Deployment to production

Benefits

* Faster delivery * Fewer errors * Improved Quality

Example

of companies uses: < * Github + Jenkins + Docker + AWS

Cloud Storage Technologies

Types Comparison

Type	Structure	Usecase
Object	unstructured	Images, videos
Block	Fixed-size blocks	Databases, OS
File	Hierarchical	Shared files

Diagram

* Object storage :

[File + metadata]

* Block Storage

[Block 1] [Block 2] [Block 3]

* File Storage

/root /folder /file.txt

Usecases

* Object → AWS S3 (Backup, media)

* Block → Databases

* File → Network shared drives

Industry Adoption

Media Companies → object storage

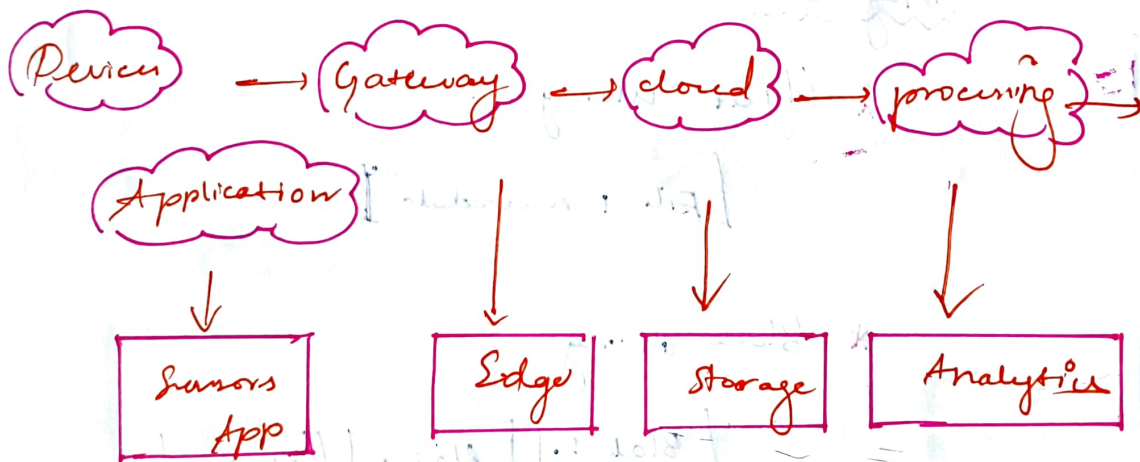
Enterprises → hybrid usage

Example

* Netflix uses object storage for video content.

IoT and cloud computing

IoT — cloud Architecture



Benefits

→ Real time data processing

→ scalability

→ Remote monitoring

Challenges:

- Security
- Data Overload
- Connectivity Issues

Applications

- smart homes
- smart cities
- Healthcare monitoring

Example

→ Smart EV system:

- * Sensors collect battery data
 - * cloud analyzes performance
 - * mobile app shows insights.
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